

# Summarizer model

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## Abstract

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This project presents an **Automated Content Summarization System** designed to process long and complex documents. The system's core function is to automatically analyze large volumes of text, **identify the most critical concepts and essential facts**, and generate concise, high-fidelity summaries. The system is built around a powerful, fine-tuned text processing model capable of handling very long documents. **This model is supported by an automated n8n workflow, which is responsible for the efficient data collection and preparation pipeline, from finding source documents to delivering the processed text to the model.** This combination ensures that the entire process, from source ingestion to final summary generation, is efficient and fully automatic.

## 1. Model Overview and Purpose

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The summarizer model is a central part of the system, designed to transform long and dense educational materials, such as books or research papers, into easily digestible summaries. The model's primary function is to perform high-fidelity extraction—it carefully reads the content and pulls out the essential scientific facts and key takeaways. This structured summary then acts as a “script” for further educational tools. The design is specifically focused on helping students, like those with dyslexia or visual learners, who can be overwhelmed by large blocks of text.

## 2. Technical Specifications

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The model is built on the Longformer Encoder-Decoder (LED) architecture. This is a special type of model that can handle much longer texts than older models, which is necessary for processing full educational documents.

| Specification     | Detail                           | Purpose                                                                                    |
|-------------------|----------------------------------|--------------------------------------------------------------------------------------------|
| Base Model        | allenai/led-base-                | The powerful foundation model designed for long-document processing.                       |
| Architecture      | Longformer Encoder-Decoder (LED) | Allows the model to read and understand very long texts.                                   |
| Max Input Length  | 4096 tokens                      | The maximum size of the text chunk the model processes at one time.                        |
| Max Target Length | 500 tokens                       | The maximum length of the summary the model will create.                                   |
| Evaluation Metric | ROUGE                            | A standard way to measure how good the generated summary is compared to a perfect summary. |

### 3. Implementation and Training Analysis

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The model was customized and trained to perform its task with high accuracy on educational content.

### 3.1. Data Handling and Preprocessing

| Data Aspect          | Detail                                                                 | Implication                                                                                                         |
|----------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Data Source          | CSV file (data_summarization.csv)                                      | The model learns from a collection of educational texts and their corresponding summaries.                          |
| Data Cleaning        | Removes rows with corrupted characters (“)                             | Ensures the model only trains on clean, readable text.                                                              |
| Splitting Strategy   | 80% Training/Validation, 20% Testing                                   | A large portion is used for learning, and a dedicated portion is saved to measure final performance.                |
| Tokenization         | Input text is limited to 4096 tokens, summary to 500 tokens.           | Converts text into a numerical format the model can understand, while managing memory.                              |
| LED Global Attention | The first token of the input is marked with a special attention value. | This is a crucial step that helps the model keep the entire context of the long document in mind while summarizing. |

### 3.2. Training Configuration

The model was fine-tuned using the following settings to optimize its performance:

| Training Parameter     | Value                 | Rationale/Implication                                                                                    |
|------------------------|-----------------------|----------------------------------------------------------------------------------------------------------|
| Epochs                 | 3                     | The model goes through the entire training dataset three times to learn.                                 |
| Per Device Batch Size  | 1                     | Indicates memory constraints, common when working with large models like LED.                            |
| Gradient Accumulation  | 8                     | This technique effectively makes the batch size 8, which is important for stable and effective training. |
| Gradient Checkpointing | Enabled               | Reduces the amount of memory needed during training, allowing for larger models and texts.               |
| Mixed Precision        | Disabled (fp16=False) | Suggests the training was done using standard precision, possibly for stability.                         |

## 4. Inference and Usage

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Once trained, the model is used to generate summaries. The generation process is carefully controlled to ensure high-quality output.

- **Generation Method:** The system uses Beam Search with a beam size of 4. This method explores the 4 most likely words at each step to build the best possible summary, resulting in a more fluent and accurate final text.
- **Length Control:** The summary length is capped at the Max Target Length of 500 tokens.
- **Early Stopping:** The generation process stops as soon as a complete summary is formed, saving time and resources.
- **Global Attention:** The special attention to the first token is applied again during generation to ensure the summary is coherent with the entire input document.

## 5. Automated Agent Workflow (n8n)

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The entire data collection and preparation process is automated using an agent workflow built with n8n. This workflow connects the steps—from finding the source material and extracting the raw text to delivering the data to the summarization model

—into a single, automatic pipeline. The n8n workflow's primary role is to manage the data pipeline, ensuring a continuous and efficient supply of processed data for the core summarization model.

## 5.1. Workflow Overview

The n8n workflow is designed to automatically ingest and process educational content from online sources. It is triggered manually, then systematically finds, downloads, extracts text from, and summarizes PDF documents, finally storing the structured results in a Google Sheet.

## 5.2. Step-by-Step Process

The workflow executes the following sequence of actions:

1. **Trigger:** The process is started by a manual execution command.
2. **Web Scraping:** An initial request is sent to a specific web page to get its content.
3. **Link Extraction:** A small piece of code searches the web page content to find all links that point to PDF documents.
4. **Iteration Loop:** The system starts a loop to process each discovered PDF link one by one.
5. **PDF Processing:** Inside the loop, the PDF is downloaded, and its text content is extracted.
6. **Summarization Request:** The extracted text is sent to a powerful Large Language Model (LLM), such as Gemini 2.5 Flash. The LLM is given a specific instruction to act as an expert summarizer, identify sections, and return the results in a structured JSON format.
7. **Output Processing:** A final code block cleans and checks the LLM's JSON output to ensure the data is correctly formatted.
8. **Data Storage:** The final structured data (including the title, summary, and original text of each section) is saved by appending a new row to a Google Sheet for easy storage and future use.

## 6. Conclusion

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The **Automated Content Summarization System** successfully combines a specialized long-context summarization model with a robust automation workflow. By fine-tuning the powerful LED model and **integrating it with an automated n8n pipeline that manages the data collection and preparation**, the project delivers an efficient and highly effective tool for information processing. This system significantly reduces the effort required to process complex materials by automatically extracting and synthesizing core concepts.